



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1 2024/2025

LAB 4 - ARP and Switch Table

SECR 1213 – NETWORK COMMUNICATIONS

SECTION 12

LECTURER: MR. FIROZ BIN YUSUF PATEL DAWOODI

NAME	MATRIC NUMBER
NUR FIRZANA BINTI BADRUS HISHAM	A23CS0156

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

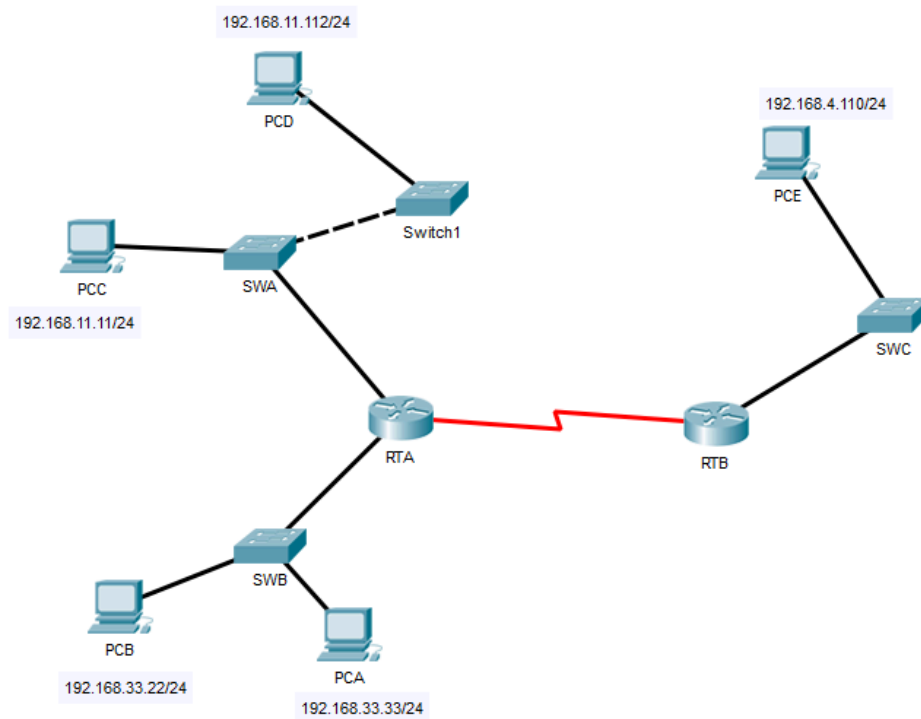


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address          Age (min)  Hardware Addr  Type   Interface
Internet 192.168.11.1          -         0002.4A00.0E91  ARPA   FastEthernet1/0
Internet 192.168.33.1          -         000C.CF0C.593A  ARPA   FastEthernet0/0
```

RTA Router

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address          Age (min)  Hardware Addr  Type   Interface
Internet 192.168.4.1          -         0001.977A.B614  ARPA   FastEthernet0/0
```

RTB Router

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0002.4a00.0e91    DYNAMIC     Fa0/1
1       000c.8546.7d85    DYNAMIC     Fa1/1
-
```

SWA switch

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a   DYNAMIC Fa0/1
```

SWB Switch

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614   DYNAMIC Fa0/1
```

SWC Switch

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCA

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCB

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
C:\>arp -a  
No ARP Entries Found  
C:\>
```

PCE

- e. What are your thoughts on the results?

The results show there is no entry for all PCs because all of the PCs didn't receive any ARP request yet. There will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.
- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time=2ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746       dynamic
```

PCA

- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06       dynamic

C:\>|
```

PCB

- In the Command prompt of PCC, PCD and PCE, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

Step 2: Generate traffic between PCC to all other PCs except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time=2ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
```

(Ping to PCB)

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic
```

(Ping to PCD)

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=2ms TTL=126
Reply from 192.168.4.110: bytes=32 time=3ms TTL=126
Reply from 192.168.4.110: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 12ms, Average = 5ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic
```

(Ping to PCE)

- e. Discuss the results you got from all the commands on PCC.
- The ARP table on PCC remains empty until it pings to PCB, PCD and PCE. No ARP entries are present to these ping operations. The ARP table moves into only after PCB, PCD and PCE receive ARP requests from PCC and respond to the subsequent ping messages.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.11.11 6 00D0.D39A.C0D9 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
Internet 192.168.33.22 6 0060.47EA.A746 ARPA FastEthernet0/0
```

RTA Router

- g. At Router RTB, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
Internet 192.168.4.110 5 0060.702D.7C08 ARPA FastEthernet0/0
```

RTB Router

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.4a00.0e91    DYNAMIC Fa0/1
1       000c.8546.7d85    DYNAMIC Fa1/1
.
```

SWA Switch

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a    DYNAMIC Fa0/1
.
```

SWB Switch

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0001.977a.b614    DYNAMIC Fa0/1
SWC#
```

SWC Switch

```
Switch>enable
Switch#show arp

Switch#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0005.5e95.187b    DYNAMIC Fa0/1
Switch#
```

Switch 1

- d. Do switches use an arp table? (Y/N)
Yes.
- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetCom syllabi, it is just for knowledge.*
Multilayer switches use ARP tables because they need to resolve IP addresses into MAC addresses to route packets between subnets or VLANs. This functionality bridges the gap between Layer 2 switching and Layer 3 routing, making ARP tables essential for efficient and accurate packet delivery.
- f. What information does the command **show mac-address-table** give?
To view the MAC address table entries on a switch, the DYNAMIC refers to MAC addresses that the switch has automatically learned during its operation.

Part 3: Attach wireless lab results.

In this part, you will use the Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot shows the Cisco Packet Tracer interface. On the left, a network topology is displayed with a central switch S1 connected to three PCs (PC1, PC2, PC3) and a wireless router WRS2. S1 has subinterfaces G0/0.10, G0/0.20, and G0/0.88. WRS2 is connected to S1 via F0/7 and F0/18. PC1 is connected to S1 via F0/11, PC2 via F0/18, and PC3 via F0/18. The IP addresses for the PCs are 172.17.10.21/24 (VLAN 10), 172.17.20.22/24 (VLAN 20), and 172.17.40.100/24. The WRS2 interface F0/18 has IP 172.17.88.25/24 (VLAN 88). The PT Activity window on the right shows the 'Configuring Wireless LAN Access' task. The Addressing Table is as follows:

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

The PT Activity window also shows 'Time Elapsed: 00:12:52' and 'Completion: 100/100'.

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

The screenshot shows the Cisco Packet Tracer interface with the same network topology as the previous image. The PT Activity window on the right shows the 'Configuring Wireless LAN Access' task with the following steps:

- Step 1: Connect the Internet interface of WRS2 to S1.**
Connect the WRS2 Internet interface to the S1 F0/7 interface.
- Step 2: Configure the Internet connection type.**
 - Click WRS2 > GUI tab.
 - Set the Internet Connection type to Static IP.
 - Configure the IP addressing according to the Addressing Table.
- Step 3: Configure the network setup.**
 - Scroll down to **Network Setup**. For the Router IP option, set the IP address to 172.17.40.1 and the subnet mask to 255.255.255.0.
 - Enable the DHCP server.
 - Scroll to the bottom of the page and click **Save Settings**.
- Step 4: Configure wireless access and security.**
 - At the top of the window, click **Wireless**. Set the Network Mode to **Wireless-N Only** and change the SSID to **WRS_LAN**.
 - Disable **SSID Broadcast** and click **Save Settings**.
 - Click the **Wireless Security** option.
 - Change the **Security Mode** from Disabled to **WPA2 Personal**.
 - Configure cisco123 as the passphrase.
 - Scroll to the bottom of the page and click **Save Settings**.

The PT Activity window also shows 'Time Elapsed: 00:25:54' and 'Completion: 100/100'.